ago, initiating life on that planet? Or, even more speculative, could life on Earth have arisen by such a transfer from Mars? Might the two planets have regularly exchanged life-forms for hundreds of millions of years? The notion might be testable. If we were to discover life on Mars and found it very similar to life on Earth—and if, as well, we were sure it wasn't microbial contamination that we ourselves had introduced in the course of our explorations—the proposition that life was long ago transferred across interplanetary space would have to be taken seriously.

IT WAS ONCE THOUGHT that life is abundant on Mars. Even the dour and skeptical astronomer Simon Newcomb (in his *Astronomy for Everybody*, which went through many editions in the early decades of this century and was the astronomy text of my childhood) concluded, "There appears to be life on the planet Mars. A few years ago this statement was commonly regarded as fantastic. Now it is commonly accepted." Not "intelligent human life," he was quick to add, but green plants. However, we have now been to Mars and looked for plants—as well as animals, microbes, and intelligent beings. Even if the other forms were absent, we might have imagined, as in Earth's deserts today, and as on Earth for almost all its history, abundant microbial life.

The "life detection" experiments on *Viking* were designed to detect only a certain subset of conceivable biologies; they were biased to find the kind of life about which we know. It would have been foolish to send instruments that could not even detect life on Earth. They were exquisitely sensitive, able to find microbes in the most unpromising, arid deserts and wastelands on Earth.

One experiment measured the gases exchanged between Martian soil and the Martian atmosphere in the presence of organic matter from Earth. A second brought a wide variety of organic foodstuffs marked by a radioactive tracer to see if there were bugs in the Martian soil who ate the food and oxidized it to radioactive carbon dioxide. A third experiment introduced radioactive carbon dioxide (and carbon monoxide) to the Martian soil to see if any of it was